Status	Finished
Started	Sunday, 25 October 2026, 9:05 PM
Completed	Sunday, 25 October 2026, 9:11 PM
Duration	5 mins 24 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Correct

Mark 1.00 out of 1.00

In which situation is linear search more efficient than binary search?

a. When the list is large and sorted
b. When the list is small and unsorted ✓
c. When the list is small and sorted
d. When the list is large and unsorted

Question **2**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. n-1
b. log n
c. n/2
d. n ✓

Question **3**

Correct

Mark 1.00 out of 1.00

Which of the following is not the required condition for a binary search algorithm?

a. Number values should only be present

Number values should only be present

b. There should be direct access to the middle element in any sublist
c. There must be a mechanism to delete and/or insert elements in the list ✓
d. The list must be sorted

Question **4**

Correct

Mark 1.00 out of 1.00

During a binary search, what happens if the target element matches the middle element?

- a. The search ends successfully ✓
- b. The search continues in the left sublist
- c. The search continues in the right sublist
- d. The list is sorted

Question **5**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the left sublist
- b. In the right sublist ✓
- c. At the beginning of the list
- d. In the middle of the list

Question **6**

Incorrect

Mark 0.00 out of 1.00

What type of search would be most appropriate for finding an element in a list that is frequently updated?

- a. Hash search ✗
- b. Binary search
- c. Linear search
- d. Interpolation search

Question **7**

Correct

Mark 1.00 out of 1.00

What happens when the element is found in linear search?

- a. The search continues until the end of the list
- b. The search starts over from the beginning
- c. The search stops immediately ✓
- d. The search backtracks to find duplicate elements

Question **8**

Correct

Mark 1.00 out of 1.00

In a linear search, how many comparisons are made in the worst-case scenario to find an element in a list of size n?

a. 1
b. n ✓
c. n/2
d. log n

Question **9**

Correct

Mark 1.00 out of 1.00

In binary search, how is the middle element determined?

a. By starting from the first element
b. By comparing each element sequentially
c. By dividing the list length by two ✓
d. By using a hash function

Question **10**

Correct

Mark 1.00 out of 1.00

What is the time complexity of binary search in the worst case?

a. O(n log n)
b. O(1)
c. O(log n) ✓
d. O👎

Question **11**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. There must be a mechanism to access middle element directly
b. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
c. Binary search algorithm is not efficient when the data elements more than 1500 ✓
d. Must use a sorted array

Question **12**

Correct

Mark 1.00 out of 1.00

What is the first step in binary search?

a. Sort the list
b. Compare the target element with the middle element in the list ✓
c. Divide the list into two equal parts
d. Compare the target element with the first element in the list

Question **13**

Correct

Mark 1.00 out of 1.00

Which of the following best describes the process of a linear search?

a. Skipping every second element
b. Checking each element sequentially ✓
c. Dividing the list in half repeatedly
d. Sorting the list before searching

Question **14**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

a. Deleting elements from a list
b. Determining whether an element is present in a list ✓
c. Sorting elements in a list
d. Inserting elements into a list

Question **15**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It does not work with negative numbers
b. It requires the list to be sorted ✓
c. It can only be applied to large lists
d. It is slower than linear search for small lists